

MACHINES

When you solve a difficult math problem, you are working hard, but this is not what scientists call work! In science, work is only done when something is moved—for example, lifted or turned around. Machines make it easier to move things, so less effort is needed to do a job. Bicycles help us move quickly. Like most machines, bicycles are made up of a number of small, simple machines. Wheels, axles, pulleys, gears, levers, slopes, and screws are all common simple machines.



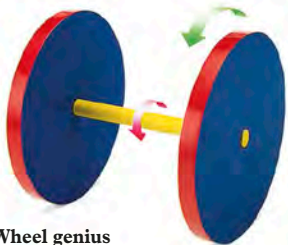
Forced apart

Wedges are simple machines that split things open. When a force hammers a wedge into a block of wood, the wood is pushed apart. A wedge is a kind of slope.



Easy does it

Levers make it easier to move things. A seesaw is a kind of lever. It balances on a point called a fulcrum. The girl's weight moves the box up. Moving the fulcrum closer to the box will make it easier to lift.



Wheel genius

The wheel is one of our most important machines. Two wheels can be joined together by a pole called an axle. A small movement of an axle makes a big turn of a wheel.



Saddle sore?

It is possible to ride a long way on a bike—between 1922 and 1973, a Scot named Tommy Chambers cycled 799,405 miles (1,286,517 kilometers)!

Cables link the brake levers to the brake blocks. Pulling the brake levers makes the brake blocks stop the moving wheels.

Brake blocks press against the wheel. They use friction to stop the wheel from moving.

The wheel spins, or rotates, around an axle.

